ECE 2305 – Introduction to C Programming
Programming Tutorial 16
FOR Structure

Write a C++ program that uses nested **`for`** loops to generate a multiplication table from 1 × 1 to 10 × 10. Use the "tab" escape sequence (**`"\t"`**) to format the table properly.

Input

```
// ECE 2305 - Tutorial 16

#include <iostream>
using namespace std;

int main() {
    cout << "Multiplication Table (1-10)" << endl;

    cout << "\t";
    for (int i = 1; i <= 10; i++) {
        cout << i << "\t";
    }
    cout << "\n----------------------------------------------------------------
----------------------\n";

    for (int i = 1; i <= 10; i++) {
        cout << i << "|\t";
        for (int j = 1; j <= 10; j++) {
            cout << (i * j) << "\t";
        }
        cout << endl;
    }

    system("pause");
    return 0;
}
```

Output

```
C:\Users\anmal\Desktop\ECE
Multiplication Table (1-10)
        1       2       3       4       5       6       7       8       9       10
-------------------------------------------------------------------------------------
1|      1       2       3       4       5       6       7       8       9       10
2|      2       4       6       8       10      12      14      16      18      20
3|      3       6       9       12      15      18      21      24      27      30
4|      4       8       12      16      20      24      28      32      36      40
5|      5       10      15      20      25      30      35      40      45      50
6|      6       12      18      24      30      36      42      48      54      60
7|      7       14      21      28      35      42      49      56      63      70
8|      8       16      24      32      40      48      56      64      72      80
9|      9       18      27      36      45      54      63      72      81      90
10|     10      20      30      40      50      60      70      80      90      100
Press any key to continue . . .
```